

Study Notes: Introduction to Python

Programming --- #### **1. Overview of

Python** * **Definition:** Python is a

**high-level, interpreted programming

language**. * **Key Characteristics:**

Known for being simple, easy to learn, and

highly readable. * **History:** Created

by **Guido van Rossum** and first released

in **1991**. #### **2. Why Python is

Popular** * **Simple Syntax:** Its clean

structure makes it beginner-friendly and

easier to write compared to other

languages. * **Powerful & Flexible:** It

can handle a wide variety of tasks, from

simple scripts to complex systems. *

Large Ecosystem: It boasts a vast

collection of **libraries**, allowing

developers to build applications quickly

and efficiently. #### **3. Programming

Paradigms** Python supports multiple

programming styles, giving developers

flexibility: * **Object-Oriented

Programming (OOP):** Focuses on "objects"

that contain data and code. *

Functional Programming: Focuses on the

use of functions. * **Procedural

Programming:** Focuses on a step-by-step

list of instructions. #### **4. Major Use

Cases** Python is widely utilized across

many technical fields: * **Web

Development** * **Artificial

Intelligence (AI) ** * **Data Science** *
Machine Learning * **Automation** *
Software Development #### **5.

Frameworks in Web Development** *

Frameworks provide pre-written code to
speed up development. * **Popular

Examples:** Flask and Django (Note: The
transcript mentions "Mongo," but
typically, **Django** is the standard
framework paired with Flask in
professional web development). *

Application: In this course, we will
use **Flask** to develop AI nodes and
generate websites. --- #### **Exam Cheat

Sheet (Key Terms)** / Term / Definition /
/ :--- / :--- / / **High-level Language**
/ A language designed to be easy for
humans to read and write. / /

Interpreted / Code is executed line-
by-line rather than compiled all at once.

/ / **Syntax** / The set of rules that
defines how a program is written (Python's
is considered "simple"). / / **Library** /

A collection of pre-written code used to
perform common tasks efficiently. / ---

*Study Tip: When preparing for exams,
focus on why Python is preferred over
other languages (simplicity + massive
library support) and be able to list at
least three areas where Python is used.*